# MSG No. 207.42 $P4'32'$ [ Type III, cubic ]

Table 1: Wyckoff site: `1a`, site symmetry: `4'32'`

| No. | position | mapping |
|---|---|---|
| 1 | [0, 0, 0] | `[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]` |

Table 2: Wyckoff site: `1b`, site symmetry: `4'32'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]` |

Table 3: Wyckoff site: `3c`, site symmetry: `4'2.2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | `[1,2,3,4,13,14,21,22]` |
| 2 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | `[5,8,10,12,17,18,19,20]` |
| 3 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | `[6,7,9,11,15,16,23,24]` |

Table 4: Wyckoff site: `3d`, site symmetry: `4'2.2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, 0\right]$ | `[1,2,3,4,13,14,21,22]` |
| 2 | $\left[0, \frac{1}{2}, 0\right]$ | `[5,8,10,12,17,18,19,20]` |
| 3 | $\left[0, 0, \frac{1}{2}\right]$ | `[6,7,9,11,15,16,23,24]` |

Table 5: Wyckoff site: `6e`, site symmetry: `4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, 0, 0]$ | `[1,2,13,14]` |
| 2 | $[-x, 0, 0]$ | `[3,4,21,22]` |
| 3 | $[0, x, 0]$ | `[5,12,17,19]` |
| 4 | $[0, 0, x]$ | `[6,9,16,23]` |
| 5 | $[0, 0, -x]$ | `[7,11,15,24]` |
| 6 | $[0, -x, 0]$ | `[8,10,18,20]` |

Table 6: Wyckoff site: `6f`, site symmetry: `4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, \frac{1}{2}, \frac{1}{2}\right]$ | `[1,2,13,14]` |
| 2 | $\left[-x, \frac{1}{2}, \frac{1}{2}\right]$ | `[3,4,21,22]` |
| 3 | $\left[\frac{1}{2}, x, \frac{1}{2}\right]$ | `[5,12,17,19]` |
| 4 | $\left[\frac{1}{2}, \frac{1}{2}, x\right]$ | `[6,9,16,23]` |
| 5 | $\left[\frac{1}{2}, \frac{1}{2}, -x\right]$ | `[7,11,15,24]` |

*continued ...*

Table 6

| No. | position | mapping |
|---|---|---|
| 6 | $\left[\frac{1}{2}, -x, \frac{1}{2}\right]$ | [8,10,18,20] |

Table 7: Wyckoff site: 8g, site symmetry: .3.

| No. | position | mapping |
|---|---|---|
| 1 | $[x, x, x]$ | [1,5,6] |
| 2 | $[x, -x, -x]$ | [2,10,11] |
| 3 | $[-x, x, -x]$ | [3,7,12] |
| 4 | $[-x, -x, x]$ | [4,8,9] |
| 5 | $[x, -x, x]$ | [13,18,23] |
| 6 | $[x, x, -x]$ | [14,15,19] |
| 7 | $[-x, x, x]$ | [16,17,21] |
| 8 | $[-x, -x, -x]$ | [20,22,24] |

Table 8: Wyckoff site: 12h, site symmetry: 2..

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, \frac{1}{2}, 0\right]$ | [1,2] |
| 2 | $\left[-x, \frac{1}{2}, 0\right]$ | [3,4] |
| 3 | $\left[0, x, \frac{1}{2}\right]$ | [5,12] |
| 4 | $\left[\frac{1}{2}, 0, x\right]$ | [6,9] |
| 5 | $\left[\frac{1}{2}, 0, -x\right]$ | [7,11] |
| 6 | $\left[0, -x, \frac{1}{2}\right]$ | [8,10] |
| 7 | $\left[x, 0, \frac{1}{2}\right]$ | [13,14] |
| 8 | $\left[0, \frac{1}{2}, -x\right]$ | [15,24] |
| 9 | $\left[0, \frac{1}{2}, x\right]$ | [16,23] |
| 10 | $\left[\frac{1}{2}, x, 0\right]$ | [17,19] |
| 11 | $\left[\frac{1}{2}, -x, 0\right]$ | [18,20] |
| 12 | $\left[-x, 0, \frac{1}{2}\right]$ | [21,22] |

Table 9: Wyckoff site: 12i, site symmetry: ..2'

| No. | position | mapping |
|---|---|---|
| 1 | $[0, y, y]$ | [1,21] |
| 2 | $[0, -y, -y]$ | [2,22] |
| 3 | $[0, y, -y]$ | [3,14] |
| 4 | $[0, -y, y]$ | [4,13] |
| 5 | $[y, 0, y]$ | [5,18] |
| 6 | $[y, y, 0]$ | [6,15] |
| 7 | $[-y, y, 0]$ | [7,16] |
| 8 | $[-y, 0, y]$ | [8,17] |
| 9 | $[-y, -y, 0]$ | [9,24] |
| 10 | $[y, 0, -y]$ | [10,19] |

*continued ...*

Table 9

| No. | position | mapping |
|---|---|---|
| 11 | $[y, -y, 0]$ | `[11,23]` |
| 12 | $[-y, 0, -y]$ | `[12,20]` |

Table 10: Wyckoff site: `12j`, site symmetry: `..2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, y, y\right]$ | `[1,21]` |
| 2 | $\left[\frac{1}{2}, -y, -y\right]$ | `[2,22]` |
| 3 | $\left[\frac{1}{2}, y, -y\right]$ | `[3,14]` |
| 4 | $\left[\frac{1}{2}, -y, y\right]$ | `[4,13]` |
| 5 | $\left[y, \frac{1}{2}, y\right]$ | `[5,18]` |
| 6 | $\left[y, y, \frac{1}{2}\right]$ | `[6,15]` |
| 7 | $\left[-y, y, \frac{1}{2}\right]$ | `[7,16]` |
| 8 | $\left[-y, \frac{1}{2}, y\right]$ | `[8,17]` |
| 9 | $\left[-y, -y, \frac{1}{2}\right]$ | `[9,24]` |
| 10 | $\left[y, \frac{1}{2}, -y\right]$ | `[10,19]` |
| 11 | $\left[y, -y, \frac{1}{2}\right]$ | `[11,23]` |
| 12 | $\left[-y, \frac{1}{2}, -y\right]$ | `[12,20]` |

Table 11: Wyckoff site: `24k`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, y, z]$ | `[1]` |
| 2 | $[x, -y, -z]$ | `[2]` |
| 3 | $[-x, y, -z]$ | `[3]` |
| 4 | $[-x, -y, z]$ | `[4]` |
| 5 | $[z, x, y]$ | `[5]` |
| 6 | $[y, z, x]$ | `[6]` |
| 7 | $[-y, z, -x]$ | `[7]` |
| 8 | $[-z, -x, y]$ | `[8]` |
| 9 | $[-y, -z, x]$ | `[9]` |
| 10 | $[z, -x, -y]$ | `[10]` |
| 11 | $[y, -z, -x]$ | `[11]` |
| 12 | $[-z, x, -y]$ | `[12]` |
| 13 | $[x, -z, y]$ | `[13]` |
| 14 | $[x, z, -y]$ | `[14]` |
| 15 | $[z, y, -x]$ | `[15]` |
| 16 | $[-z, y, x]$ | `[16]` |
| 17 | $[-y, x, z]$ | `[17]` |
| 18 | $[y, -x, z]$ | `[18]` |
| 19 | $[y, x, -z]$ | `[19]` |
| 20 | $[-y, -x, -z]$ | `[20]` |
| 21 | $[-x, z, y]$ | `[21]` |
| 22 | $[-x, -z, -y]$ | `[22]` |

*continued ...*

Table 11

| No. | position | mapping |
|---|---|---|
| 23 | $[z, -y, x]$ | [23] |
| 24 | $[-z, -y, -x]$ | [24] |